

Type 'TWO' System: $G(s) = \frac{k}{s^2(1+sT_1)}$

Let, $s = j\omega$,

$$\therefore G(j\omega) = \frac{k}{(j\omega)^2(1+j\omega T_1)}$$

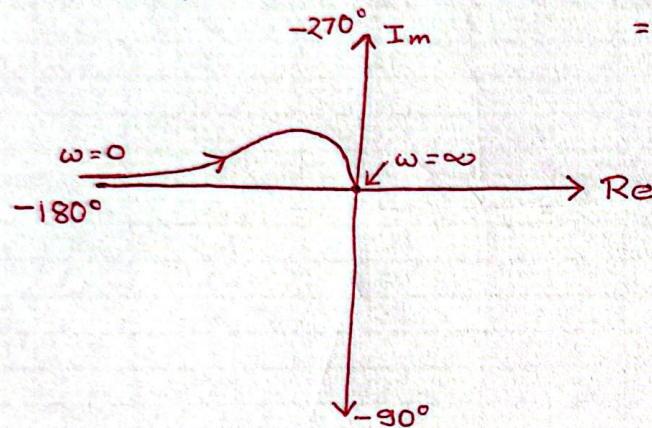
As, $(j\omega)^2 = -\omega^2$, $G(j\omega) = \frac{k}{-\omega^2(1+j\omega T_1)}$

Magnitude, $|G(j\omega)| = \frac{k}{\omega^2 \sqrt{1+(\omega T_1)^2}}$

Phase, $\angle G(j\omega) = -180^\circ - \tan^{-1}(\omega T_1)$

$\omega = 0$; $|G(j0)| = \infty$ $\angle G(j0) = -180^\circ$

$\omega = \infty$; $|G(j\infty)| = 0$ $\angle G(j\infty) = -180^\circ - 90^\circ = -270^\circ$



Magnitude rule: $\left| \frac{X}{Y} \right| = \frac{|X|}{|Y|}$

$|k| = k$

$|- \omega^2| = \omega^2$

$|a+jb| = \sqrt{a^2+b^2}$

$|1+j\omega T_1| = \sqrt{1+(\omega T_1)^2}$

Phase, $\angle \frac{A}{B} = \angle A - \angle B$

$\angle k = 0^\circ$

Phase of $j\omega = +90^\circ$

so square of $j\omega = +180^\circ$

$\angle (j\omega)^2 = +180^\circ$

$\angle a+jb = \tan^{-1} \frac{b}{a}$

$\therefore \angle 1+j\omega T_1 = \tan^{-1} \omega T_1$

Q. Can it cut Real axis? or imaginary axis?

Introduction of additional pole: $G(s) = \frac{k}{s^2(1+sT_1)(1+sT_2)}$

$$s = j\omega ; \quad G(j\omega) = \frac{k}{(j\omega)^2(1+j\omega T_1)(1+j\omega T_2)}$$

$$\text{So, } G(j\omega) = \frac{k}{-\omega^2 \sqrt{1+(\omega T_1)^2} \sqrt{1+(\omega T_2)^2}} \angle -180^\circ - \tan^{-1} \omega T_1 - \tan^{-1} \omega T_2$$

$$\omega = 0 ; \quad |G(j\omega)| = \infty \quad \angle G(j\omega) = -180^\circ$$

$$\omega = \infty ; \quad |G(j\omega)| = 0^\circ \quad \angle G(j\omega) = -360^\circ$$

Separating real and imaginary part of $G(j\omega)$.

$$\text{Re}\{G(j\omega)\} = -\omega^2 + \omega^4 T_1 T_2, \quad \text{Im}\{G(j\omega)\} = -\omega^3 (T_1 + T_2)$$

Equating the real part to '0'. $-\omega^2 + \omega^4 T_1 T_2 = 0$

$$\Rightarrow \omega^2 = \omega^4 T_1 T_2$$

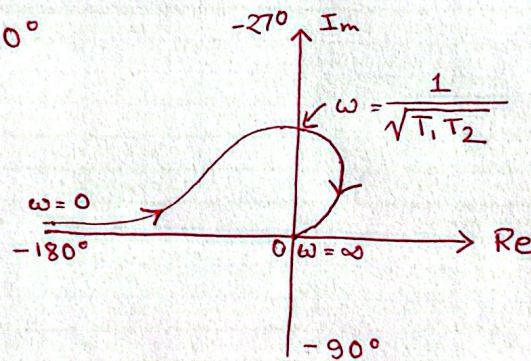
$$\Rightarrow \omega^2 = \frac{1}{T_1 T_2}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{T_1 T_2}}$$

$$\angle G(j\omega) = \angle -180^\circ - \tan^{-1} \sqrt{\frac{T_1}{T_2}} - \tan^{-1} \sqrt{\frac{T_2}{T_1}} \quad [\text{We know, } \tan^{-1} a + \tan^{-1} \frac{1}{a} = 90^\circ]$$

$$= \angle -180^\circ - 90^\circ$$

$$= \angle -270^\circ$$



Some standard type function:

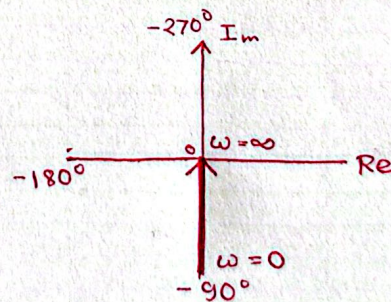
$$1. \quad G(s) = \frac{1}{s}$$

$$G(j\omega) = \frac{1}{j\omega}$$

$$G(j\omega) = \frac{1}{\omega} \angle -\tan^{-1} \omega = \frac{1}{\omega} \angle -90^\circ$$

$$\omega = 0 ; \quad |G(j0)| = \infty$$

$$\omega = \infty ; \quad |G(j\infty)| = 0$$



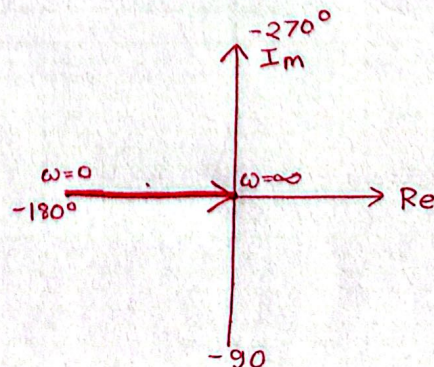
$$2. \quad G(s) = \frac{1}{s^2}$$

$$G(j\omega) = \frac{1}{(j\omega)^2}$$

$$G(j\omega) = \frac{1}{-\omega^2} \angle -180^\circ$$

$$\omega = 0 ; \quad |G(j0)| = \infty$$

$$\omega = \infty ; \quad |G(j\infty)| = 0$$



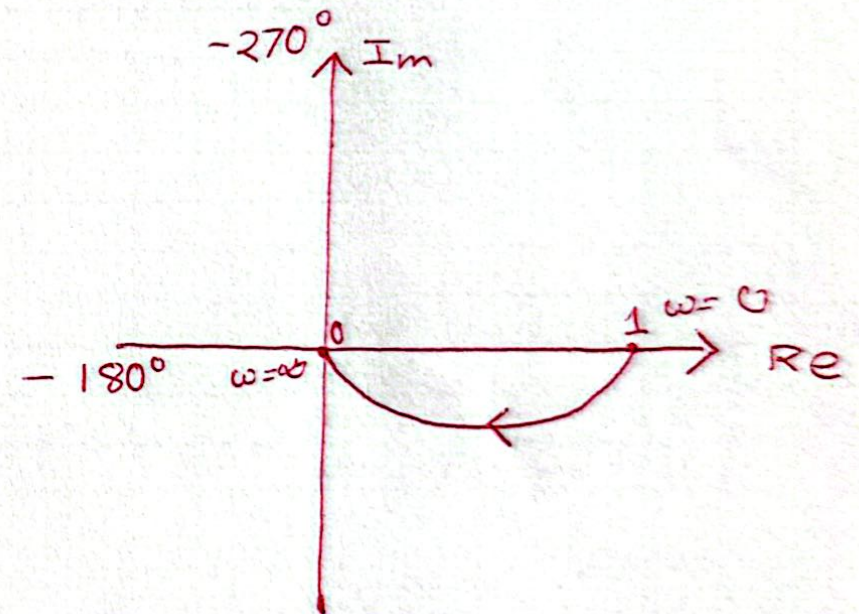
$$3. \quad G(s) = \frac{1}{1+s}$$

$$G(j\omega) = \frac{1}{1+j\omega}$$

$$G(j\omega) = \frac{1}{\sqrt{1+\omega^2}} \angle -\tan^{-1}\omega$$

$$\omega=0; \quad |G(j\omega)| = 1 \quad \angle G(j\omega) = 0^\circ$$

$$\omega=\infty; \quad |G(j\omega)| = 0 \quad \angle G(j\omega) = -90^\circ$$



Q. Why this polar plot is a curved line, why not a straight line?

Q. Sketch the polar plot for $G(s) = \frac{20}{s(s+1)(s+2)}$

$$G(j\omega) = \frac{20}{j\omega(j\omega+1)(j\omega+2)}$$

$$G(j\omega) = \frac{20}{\omega \sqrt{1+\omega^2} \sqrt{4+\omega^2}} \angle -90^\circ - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

$$\omega = 0; |G(j\omega)| = \infty \quad \angle G(j\omega) = -90^\circ$$

$$\omega = \infty; |G(j\omega)| = 0 \quad \angle G(j\omega) = -270^\circ$$

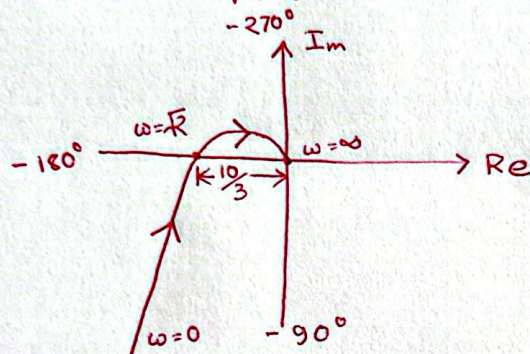
Separating the real and imaginary part of $G(j\omega)$.

$$j\omega(j\omega+1)(j\omega+2) = j\omega(-\omega^2 + j2\omega + j\omega + 2) = -j\omega^3 - 3\omega^2 + j2\omega$$

$$\operatorname{Re}\{G(j\omega)\} = -3\omega^2, \quad \operatorname{Im}\{G(j\omega)\} = -\omega^3 + 2\omega$$

Putting $\operatorname{Im}\{G(j\omega)\} = 0 \Rightarrow \omega^2 = 2 \Rightarrow \omega = \sqrt{2}$

$$\therefore |G(j\omega)| = \frac{20}{\sqrt{2} \times \sqrt{1+2} \times \sqrt{4+2}} = \frac{20}{6} = \frac{10}{3} \quad \text{and} \quad \angle G(j\omega) = \angle -90^\circ - 55^\circ - 35^\circ = \angle -180^\circ$$



Q. $G(s) = \frac{10}{s(s+1)}, \quad G(j\omega) = \frac{10}{j\omega(j\omega+1)}$

$$G(j\omega) = \frac{10}{\omega \sqrt{\omega^2+1}} \angle -90^\circ - \tan^{-1}\omega$$

$$\omega = 0; |G(j\omega)| = \infty \quad \angle G(j\omega) = -90^\circ$$

$$\omega = \infty; |G(j\omega)| = 0 \quad \angle G(j\omega) = -180^\circ$$

For getting real and imaginary part.

$$j\omega(j\omega+1) = -\omega^2 + j\omega$$

$$\operatorname{Re}\{G(j\omega)\} = 0 \quad \text{and} \quad \operatorname{Im}\{G(j\omega)\} = 0$$

$$\Rightarrow -\omega^2 = 0 \quad \Rightarrow \omega = 0$$

$$\Rightarrow \omega = 0$$

So, the polar plot intersects the real and imaginary axis at the origin only.

